# Entropy as a Projection Artifact: One-Body Operators are Entanglement-Inert on Sparse Lattices

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We characterize the operator-theoretic origin of entanglement entropy in the GeoVac Coulomb lattice, a finite spectral graph realization of the hydrogenic angular momentum eigenbasis on $S^3$. Five results are presented. First, the von Neumann entropy of any single-particle reduced density matrix derived from a *non-degenerate* ground state of a purely one-body fermionic Hamiltonian on a finite lattice is identically zero; numerically, we confirm $S_{\rm kin}/S_{\rm full} \sim 10^{-14}$ at $n_{\max} = 2, 3$ on a He model whose one-body operator is deliberately non-diagonal in the angular momentum eigenbasis (for helium's physical, diagonal one-body operator the zero is exact). All ground-state entanglement is generated by the two-body interaction $V_{ee}$, with the non-degeneracy qualifier essential: open-shell and dissociated-bond states escape the floor. Second, the holographic area-law scaling $S_n = k \log A_n$ with $A_n \propto n^4$, asserted on one-particle grounds in earlier framework papers, is rigorously recovered as a pair-counting statement: $A_n = g_n^2$, where $g_n = 2n^2$ is the shell degeneracy. The factor of four is the two-body signature. Third, in the Coulomb lattice the cusp concentrates as a single "hot node" on the $(1s, 1s)$ pair-state of the $V_{ee}$ graph; since entropy is generated exclusively by $V_{ee}$, this is also the locus of entanglement mass: an edge-resolved decomposition puts 58.5% of helium's correlation entropy on the single hottest $V_{ee}$ edge, while an equally large edge away from the cusp vertex carries none. Fourth, a previously claimed rigidity statement — that the ground-state entropy of any two-fermion closed-shell state on the Bargmann–Segal harmonic oscillator lattice interacting via *any* central two-body potential is identically zero, by Moshinsky–Talmi total-quanta conservation — is RETRACTED here (§VII A): the Moshinsky–Talmi *bracket* conserves total quanta, the potential *matrix element* does not, and the published zero was a code artifact. The corrected entropy is small but nonzero and *grows* with basis size. Fifth, Coulomb-lattice block entropy obeys a universal scaling $S_B = A\,(\tilde{w}_B/\delta_B)^{\gamma}$, where $\tilde{w}_B$ is the dimensionless $V_{ee}$ mass off-diagonal in the one-body eigenbasis and $\delta_B$ the dimensionless first $H_1$ gap; the local exponent extrapolates to $\gamma_\infty \approx 1.96$–$1.97$ in the $n_{\max} \to \infty$ limit at the large-$Z$ edge (extrapolated from $n_{\max} = 2$–$7$ on helium-like ions' physical Hamiltonian; fixed-$n_{\max}$ local slopes reach 1.91 by $Z = 100$; an earlier 1.93–1.96 was measured on a hop-augmented model one-body operator), slightly below the non-degenerate second-order Rayleigh–Schrödinger value of 2, with the residual gap attributed to multi-shell aggregation in the basis-extension limit. Entropy is slotted into the exchange-constant taxonomy as a projection-class observable: zero on the combinatorial graph (non-degenerate case), finite only after the metric embedding of $1/r_{12}$.

## I. INTRODUCTION

The kinetic and potential parts of a fermionic Hamiltonian play asymmetric roles in generating entanglement. Slater determinants exactly diagonalize one-body operators and carry only the entanglement floor mandated by antisymmetrization; all non-trivial orbital correlation arises from two-body terms. This fact is folklore in the quantum-chemistry and quantum-information literatures (see, e.g., [1, 2]), but its sharp form is rarely written out, and its consequences on *sparse, lattice-realized* Hamiltonians are seldom exhibited explicitly.

The GeoVac framework [4, 6, 7] provides a controlled testbed: the hydrogenic angular momentum eigenbasis labelled by the vertices of a finite graph on $S^3$, where the physical one-body operator $H_1$ is the diagonal hydrogenic spectrum $-Z^2/2n^2$ (exact at orbital exponent $k = Z$), and the two-body operator $V_{ee}$ is the Coulomb repulsion $1/r_{12}$ projected onto the same vertex set. Paper 26 [12] observed that a one-body term added to this Hamiltonian—a constant $+\frac{1}{16}$ hop along the graph's edges, then read as "graph Laplacian topology" and now recognized as a model term that is not part of the atom's Hamiltonian (see §III)—moves 10–39% of the model's correlation energy but carries $< 0.2\%$ of [retracted 2026-09-25: he-graph-native-hop-headline] its single-orbital entanglement entropy. That leaves open whether this asymmetry is a quantitative numerical observation specific to the basis at hand or a structural identity that holds at the operator level.

This paper resolves the question. Section II states the operator-theoretic fact: on any finite lattice, the one-body operator is entanglement-inert at the closed-shell ground state. Section III confirms the result to machine precision on He at $n_{\max} = 2, 3$. Section IV rederives the holographic area-law $S_n \propto 4 \log n$ as a pair-counting statement, fixing a mislabeling in Paper 5 [5] where the scaling was presented in one-particle language. Section V shows that the cusp localizes as a single hot node on the two-body $V_{ee}$ graph, making entropy and cusp mass spatially co-located. Section VI places entropy in the exchange-constant taxonomy of Paper 18 [8] as a projection-class observable. Section VII states two falsifiable predictions, one for the Bargmann–Segal harmonic

oscillator lattice [11] — which the present version RETRACTS — and one across the GeoVac molecular library, which stands.

*Provenance (register tiers).* The entanglement-inertness of the one-body operator (§ II) is an INTERNAL THEOREM. The harmonic-oscillator zero-entropy rigidity, previously carried at the same tier, is RETRACTED (§ VII A) — it was *measured* against a restricted evaluation object, not proved; the $S_{\rm kin}/S_{\rm full} \sim 10^{-14}$ numerical confirmation and the block-entropy exponent $\gamma_\infty \approx 1.96$–$1.97$ (a Richardson extrapolation on the physical Hamiltonian, *not* an exact value; remeasured 2026-09-25, previously 1.93–1.96 on a hop-augmented model) are MEASURED; the area-law $A_n = g_n^2$ recovery is an exact combinatorial identity (SYMBOLIC). Entropy is placed as an embedding-class OBSERVATION in the Paper 18 exchange-constant taxonomy.

## II. OPERATOR-THEORETIC RESULT

### A. Setting

Let $\mathcal{B} = \{\phi_1, \ldots, \phi_M\}$ be a finite orthonormal single-particle basis on a lattice (the vertex set of a spectral graph), and let

$$H_1 \;=\; \sum_{ij} h_{ij}\, c_i^\dagger c_j, \qquad h = h^\dagger, \tag{1}$$

be a one-body Hermitian operator on the antisymmetric Fock space generated by $\mathcal{B}$. Let $\Phi_0$ be the ground state of $H_1$ at fixed particle number $N$.

### B. Statement

**Proposition.** *If the spectrum of $h$ is non-degenerate at the Fermi level (i.e., $\Phi_0$ corresponds to a uniquely defined Slater determinant of the $N$ lowest natural orbitals of $h$), then for any subset $A \subset \mathcal{B}$ of orbitals the von Neumann entropy*

$$S_A \;=\; -\mathrm{Tr}\left[\rho_A \log \rho_A\right], \qquad \rho_A \;=\; \mathrm{Tr}_{\mathcal{B}\setminus A}\, |\Phi_0\rangle\langle\Phi_0|, \tag{2}$$

*is determined entirely by the natural-orbital occupation spectrum of $\Phi_0$ restricted to $A$, which consists of zeros and ones in the natural-orbital basis. Equivalently, in the basis that diagonalizes $h$, $\rho_A$ is a tensor product of single-mode density matrices each of which has support on a single occupation eigenvalue. In a generic basis $\mathcal{B}$, the resulting entropy is the antisymmetrization (Slater) entropy [3], and is independent of $h$.*

The content of the proposition is well known in quantum-information formulations of free-fermion systems [1]: a Slater determinant has a Gaussian correlation matrix, and its reduced density matrices are fully characterized by the projector $C = P_A^\dagger P P_A$ onto the occupied subspace, restricted to the partition $A$. The eigenvalues of $C$ are pinned to $\{0, 1\}$ in the natural basis; any positive entropy of $\rho_A$ in a different basis is a basis-projection artifact, not a physical correlation.

### C. Scope of the proposition

The non-degeneracy clause is essential. An empirical scan (EP-2j) at $n_{\max} = 3$ in the GeoVac $S^3$ basis with $N = 3, 4$ electrons shows two distinct violation modes:

*Open-shell single Slater determinant (Li, $^2S$ ground state).* The $H_1$ ground state is non-degenerate (gap $\sim 10^{-5}$ Ha) and is exactly one Slater determinant $|1s^2\, 2s\rangle$, so the proposition's premise holds in the canonical statement of Sec II. However, the spatial 1-RDM has eigenvalues $(2, 1, 0, \ldots)$, and the von Neumann entropy of the normalized 1-RDM $S = -\sum_i (n_i/N)\log(n_i/N) = -\frac{2}{3}\log\frac{2}{3} - \frac{1}{3}\log\frac{1}{3} \approx 0.637$ nats is nonzero not from $V_{ee}$ correlation but from the open-shell *antisymmetrization* of the single SD. The $V_{ee}$ correlation contribution is the small differential $S_{\rm full} - S_{\rm kin} \approx 0.010$ nats at $Z = 3$. The proposition's $S_A$=0 statement applies in the *natural-orbital basis of $h$* where the open-shell SD has occupation $(1, 1, 1, 0, \ldots)$ in spin-orbital units, giving $S = 0$ trivially in that basis. In the spatial 1-RDM, the antisymmetrization entropy is the relevant floor, not zero.

*Genuine $H_1$ degeneracy (Be, $^1S$ ground state at $n_{\max} = 3$).* The bare graph $H_1$ has a 3-fold degenerate ground state from the unbroken $2s/2p$ shell degeneracy (the diagonal $-Z^2/2n^2$ does not split $2s$ from $2p$, and the $\kappa$-adjacency does not lift it in the $M_L = 0, M_S = 0$ sector). The $H_1$-only ground ensemble is mixed, the spatial 1-RDM has fractional occupations $(2, 0.76, 0.76, 0.47, 0, \ldots)$, and $S_{\rm kin} = 1.23$ nats $\neq 0$. $V_{ee}$ lifts the degeneracy and *reduces* entropy: $S_{\rm full} = 0.902 < S_{\rm kin}$. Here the proposition's premise fails outright; the floor is $S_{\rm kin} > 0$, not zero.

The He two-electron case satisfies the proposition's premise in the strictest form: $1s^2$ closed shell exhausts the non-degenerate $n = 1$ shell, the GS is a single SD, the spatial 1-RDM has eigenvalues $(2, 0, \ldots)$, and the floor is $S_{\rm kin} = 0$. Generically the proposition's spatial-basis floor $S_{\rm kin} = 0$ applies to closed-shell ground states whose electron count exhausts an entire $(n, l)$-degenerate sub-shell at the $h$-eigenbasis level. For the GeoVac $S^3$ basis at $n_{\max} \geq 2$, this restricts the strict floor to He-like atoms and to systems whose closed shells fall entirely in the lowest $1s$ sub-shell. The proposition's operator content is general; its quantitative spatial-basis realization narrows.

*Analytical degenerate perturbation theory for Be (EP-2N).* The Be case admits a clean analytical treatment. At $n_{\max} = 2$, $M_L = M_S = 0$, the $H_1$ ground subspace is exactly 3-dimensional, spanned by $\{|1s^2 2s^2\rangle, |1s^2 2p_0^2\rangle, |1s^2\,(2p_{-1}2p_{+1})_{S=0}\rangle\}$ at common

$E_{\rm kin} = -5Z^2/4$. Diagonalizing $V_{ee}$ within this subspace (a $3\times3$ degenerate-PT problem) at $Z = 4$ yields a ground state $0.98\,|2s^2\rangle - 0.14\,|2p_0^2\rangle - 0.14\,|(2p_{-1}2p_{+1})_{S=0}\rangle$ – 96% pure $2s^2$ with $\sim 2\%$ admixture each of the two $2p$-pair configurations. The analytical spatial 1-RDM has diagonal occupations $(2.000, 1.924, 0.020, 0.037, 0.020)$ in the $(1s, 2s, 2p_{-1}, 2p_0, 2p_{+1})$ basis, giving $S_{\rm full} = 0.794$ nats vs. the equal-mixture $S_{\rm kin} = 1.358$ nats. $V_{ee}$ thus acts as a *degenerate-PT lifter*: it selects the $2s^2$ configuration and reduces entropy by 42%. The full FCI calculation at $n_{\max} = 3$ (EP-2j) gives $S_{\rm full} = 0.902$ in qualitative agreement, with the additional shells contributing modest extra correlation entropy on top of the analytical degenerate-PT prediction. This sharpens the post-degeneracy floor to a calculable quantity rather than the trivial $S_{\rm kin} = 0$ of the non-degenerate case.

### D. GeoVac specialization

For the GeoVac Coulomb lattice, $\mathcal{B}$ is the angular momentum eigenbasis $\{\phi_{nlm}\}$ on $S^3$ at orbital exponent $k = Z$. In this basis the physical one-body operator is exactly diagonal,

$$H_1 \;=\; \mathrm{diag}(\varepsilon_{nlm}), \qquad \varepsilon_{nlm} = -\frac{Z^2}{2n^2}, \tag{3}$$

because the $\phi_{nlm}$ are eigenfunctions of $-\frac{1}{2}\nabla^2 - Z/r$. $H_1$ is non-degenerate at the closed-shell He ground state $(1s^2)$ for all $n_{\max} \geq 1$, so by the proposition the single-orbital entropy of the $H_1$-only ground state vanishes; here the natural orbitals coincide with $\{\phi_{nlm}\}$ and the vanishing is exact.

**[CORRECTED 2026-09-25.]** Earlier versions wrote $H_1 = \mathrm{diag}(\varepsilon_{nlm}) + \kappa L$ with $\kappa = -\frac{1}{16}$ and $L$ the GeoVac graph Laplacian—i.e. a constant $+\frac{1}{16}$ hop added along every lattice edge. That hop is not part of the atom's Hamiltonian (it has no derivation in this basis, where the physical one-body matrix has no off-diagonal elements, and the diagonal is not $\kappa$ times any graph operator), and it lowers the He energy by $\approx 49$ mHa, more than helium's whole correlation energy. The proposition is operator-general, so it holds for the hop-augmented model as well—there the natural orbitals deviate from $\{\phi_{nlm}\}$ by $O(|\kappa|/Z^2)$ and the entropy in $\mathcal{B}$ inherits a small projection contribution—but the *numbers* of Sections III, V and VII B were measured on that model. Each is re-measured on the physical Hamiltonian below; every qualitative conclusion survives, and several numbers move.

## III. NUMERICAL CONFIRMATION

We solve for the ground state of two He Hamiltonians on the $S^3$ angular momentum eigenbasis at $n_{\max} \in \{2, 3\}$:

TABLE I. One-body vs. full ground-state entropy for He on the $S^3$ angular lattice. $S_{\rm full}$ and $S_{\rm kin}$ are the single-orbital von Neumann entropies of the FCI ground states of $H_{\rm full}$ and $H_{\rm kin}$, respectively. Data from `debug/data/ep1_entropy_partition.json` [13].

| $n_{\max}$ | $E_{\rm full}$ (Ha) | $S_{\rm full}$ (nats) | $S_{\rm kin}$ (nats) | $S_{\rm kin}/S_{\rm full}$ |
|---|---|---|---|---|
| physical $H_1$ [Eq. (3)] (2026-09-25) | | | | |
| 2 | −2.8334 | 0.03352 | 0 | 0 |
| 3 | −2.8429 | 0.03772 | 0 | 0 |
| hop-augmented model (EP-1 data; not helium) | | | | |
| 2 | −2.8895 | 0.03955 | $-2.2\times10^{-16}$ | $-5.6\times10^{-15}$ |
| 3 | −2.8931 | 0.04081 | $7.8\times10^{-16}$ | $1.9\times10^{-14}$ |

- the full Hamiltonian $H_{\rm full} = H_1 + V_{ee}$, with $V_{ee}$ from the exact-rational hypergeometric Slater integral evaluator [6, 13];
- the kinetic-only Hamiltonian $H_{\rm kin} = H_1$, identical to $H_{\rm full}$ with $V_{ee} \equiv 0$.

We diagonalize each in the $M_L = 0$ singlet sector by full configuration interaction, build the spatial 1-RDM, and report its von Neumann entropy in nats.

Table I reproduces the EP-1 data file [13].

With the physical $H_1$ the one-body ground state is the single determinant $1s^2$ in the working basis, so $S_{\rm kin}$ is exactly zero, not merely at the noise floor. The hop-augmented model is the non-trivial test of the proposition: there $H_1$ is not diagonal in $\mathcal{B}$, and the entropy still vanishes to machine precision. The paragraph below describes the model rows.

The kinetic-only entropy is at the floating-point noise floor ($\sim 10^{-16}$ nats), confirming the proposition to machine precision. The full Hamiltonian's entropy ($S_{\rm full} \sim 4\times10^{-2}$ nats) is fourteen orders of magnitude larger. The natural-orbital occupations of the kinetic-only ground state at $n_{\max} = 3$ are $(2.000, 8.3\times10^{-22}, 0, 0, \ldots)$, exactly the $\{0, 1\}$ structure (here doubled by spin) predicted by the proposition. The full Hamiltonian's occupations $(1.988, 0.00896, 0.00093, 0.00093, \ldots)$ exhibit the $O(10^{-2})$ deviations from $\{0, 2\}$ (largest $2-1.988 = 0.012$) characteristic of the two-body $V_{ee}$ correlation.

The result is expected. Its value lies in making it visible on the concrete Coulomb lattice and in demonstrating that the floor is operator-theoretic, not basis-fitted.

## IV. AREA LAW AS PAIR COUNTING

### A. The Paper 5 statement

The framework synthesis paper [5] states (Section "Entropy and the second law", verbatim):

> The holographic entropy of shell $n$ is $S_n = k \ln A_n + \text{const}$, where $A_n \propto n^4$ is the "surface area" (number of plaquettes).

The supporting microstate count was not given. The shell degeneracies in Paper 0 [4] are $g_n^{\rm orb} = n^2$ (orbital, no spin) and $g_n = 2n^2$ (with spin); cumulative state count is $N(n) = n(n+1)(2n+1)/3 \sim (2/3)n^3$. Neither yields $n^4$.

### B. Pair counting on shell $n$

Define a *plaquette* on shell $n$ as an ordered pair of sites,

$$A_n \;\equiv\; g_n^2 \;=\; 4\,n^4. \tag{4}$$

Equivalently, the unordered antisymmetrized pair count is

$$\tfrac{1}{2}\,g_n(g_n-1) \;=\; n^2(2n^2-1) \;=\; 2\,n^4 + O(n^2), \tag{5}$$

and the adjacent-shell pair count is $g_n g_{n-1} = 4n^2(n-1)^2 = 4n^4 + O(n^3)$. All three two-body counts agree at leading order:

$$\log A_n \;=\; 4\,\log n \;+\; \text{const.} \tag{6}$$

A factor of four arises only at the two-body level: each electron contributes $\log g_n = 2\log n + \log 2$, and the two factors compose multiplicatively. At the one-body level, single-shell uniform entropy is $\log g_n = 2\log n + \log 2$ (off by a factor of two), and cumulative entropy $\log N(n) \sim 3\log n$ (off by a factor of 4/3). These are exhibited explicitly in the EP-3 analysis [16].

The area-law scaling $A_n \propto n^4$ asserted in Paper 5 is therefore correct, but it is a *two-body* statement. In the operator-theoretic language of Section II, the area law lives on the pair graph of the $V_{ee}$ operator, not on the single-particle vertex set of $H_1$.

### C. Reconciliation

Paper 5 is a predecessor framework synthesis with conjectural status. Its area-law claim is rigorously recovered here as a pair statement. Two consequences follow:

1. The area law cannot be derived from one-particle microstate counting on the GeoVac packing lattice. Any interpretation that assigns the $n^4$ scaling to a one-particle observable is mislabeled.

2. The information-theoretic content of the scaling is consistent with Section II: entropy on the GeoVac lattice is generated by the two-body operator, and its leading shell scaling counts pair states.

## V. CUSP AS ENTROPY CONCENTRATION

The energy-graph characterization in v2.9.1 [17] constructs the singlet pair-state graph $\{|(ab)\rangle\}$ at $n_{\max} = 3$ ($N = 31$ nodes) and $n_{\max} = 4$ ($N = 101$ nodes), with edge weights $V_{IJ} = \langle (ab)|1/r_{12}|(cd)\rangle$ computed from the exact-float hypergeometric Slater integrals [6]. Two findings of that characterization bear directly on the operator-theoretic result of Section II.

*Corrected 2026-09-25.* The characterization and the decomposition below were first run on a system that was not helium on two counts: the hop-augmented $H_1$ of §II, and edge weights $V_{IJ}$ evaluated at unit orbital exponent ($k = 1$) while $H_1$ was at $Z = 2$, i.e. $1/r_{12}$ at half strength. (The $V$ graph's *shape*—which edge is hottest, the ratios between edges—is $k$-independent, since $V_{ee} \propto k$; the entropies and the commutator are not.) The numbers below are re-measured on helium's physical Hamiltonian, $H_1$ of Eq. (3) plus $V_{ee}$ at $k = Z = 2$, with the earlier values in parentheses.

*Frobenius-mass diagonality.* In the eigenbasis of the one-body operator $H_1$, $V_{ee}$ is overwhelmingly diagonal:

$$\frac{\|\mathrm{diag}_{H_1} V_{ee}\|_F}{\|V_{ee}\|_F} \;=\; 0.945\ (n_{\max}=3),\ \ 0.926\ (n_{\max}=4), \tag{7}$$

using the unsquared Frobenius ratio (earlier, on the hop-augmented model: 0.920, 0.892). The relative commutator $\|[H_1, V_{ee}]\|_F / \|H_1\|_F = 10.6\%$ at $n_{\max} = 3$, dropping only to 9.3% at $n_{\max} = 4$ (earlier 6.1%, 5.3%, with $V_{ee}$ at half strength). The commutator does not vanish with basis size; $H_1$ and $V_{ee}$ are genuinely non-commuting. The one-body eigenbasis "almost" diagonalizes the entropy-generating operator, and the residual off-diagonal mass is the correlation. (An earlier sentence credited it with driving the "graph-native" CI from $\sim 3\%$ to 0.2%; that convergence belonged to the hop-augmented model, and with the physical $H_1$ the fixed-$k$ CI stops near 1.9%, a floor set by the basis's missing continuum.)

*Hot-node localization.* On the same pair-state graph, the $(1s, 1s)$ vertex is simultaneously the maximum-diagonal entry ($V_{(1s,1s),(1s,1s)} = 5k/8$, with the next-largest diagonal at 41.6% of it), the largest weighted hub, and the head of the largest off-diagonal edge $(1s, 1s) \leftrightarrow (1s, 2s)$. The cusp does not distribute across pair-states; it concentrates on the unique pair-state where both electrons can simultaneously occupy small-$r$ regions. Combined with Section II, this identifies the cusp not only as the energy-evaluation bottleneck [8] but as the locus of entanglement mass: entropy is generated exclusively by $V_{ee}$ (Section II) and $V_{ee}$'s mass concentrates here.

**The decomposition (2026-08-29).** This was carried as an inference until a decomposition existed; one does now. In the singlet pair-state basis the $V_{ee}$ graph's *nodes* and the CI *configurations* are the same objects, so entropy can be attributed to graph edges without a perturbative step: zero a single coupling $V_{{\rm ref},J}$ out of the dominant $(1s, 1s)$ reference node, re-solve, and record the entropy lost. For He at $Z = 2$, $n_{\max} = 3$ the single edge $(1s, 1s) \leftrightarrow (1s, 2s)$ carries 58.5% of the total $S_{\rm full} = 3.77 \times 10^{-2}$ nats (earlier 42.5% of $5.6 \times 10^{-3}$ on the model), and it is exactly the hottest off-diagonal edge of the $V_{ee}$ graph identified above. At $n_{\max} = 4$ it is the

same edge at 54.1% (earlier 40.0%), so the co-location is not a small-basis artifact.

Two features make this more than a restatement. First, the control: the largest $V_{ee}$ edge that does *not* touch the reference node, $(1s, 2p_0) \leftrightarrow (1s, 3p_0)$ with $|V|$ half the hot edge's, changes the entropy by $\sim 10^{-15}$ nats — numerically nothing. So what generates entropy is not $V_{ee}$ mass anywhere in the graph, but $V_{ee}$ mass *at the cusp vertex*. Second, the correspondence is graded, not merely top-ranked: across all edges out of the reference the entropy ordering tracks the $V_{ee}$ ordering at Spearman $\rho = +0.87$ ($+0.81$ at $n_{\max} = 4$; earlier $+0.94$, $+0.92$). It is not proportionality — the $(2s, 2s)$ edge carries 23.5% of the entropy on a coupling six times smaller than the top edge's, because the attribution weights $V/\Delta E$ rather than $V$ — but the ranking and the maximum coincide. *Tier:* MEASURED, in the He ground state at $n_{\max} = 3, 4$; the attribution is leave-one-edge-out, which is well-defined but not the unique possible decomposition.

## VI. EXCHANGE-CONSTANT TAXONOMY PLACEMENT

Paper 18 [8] classifies observables by the type of transcendental content their evaluation requires: *intrinsic* (rational graph-native quantities, e.g. Slater $F^0$ values), *calibration* (transcendental constants of the conformal projection, e.g. $\pi$ from $S^3$ Weyl densities), *embedding* (constants that enter when the graph is mapped to a continuous metric, e.g. the Coulomb $1/r_{12}$ kernel), and *flow* (renormalization-style constants, e.g. Phillips–Kleinman).

Entropy fits the framework cleanly as a *projection-class observable*:

- On the bare combinatorial graph (vertex set $\{\phi_{nlm}\}$, one-body operator $H_1$ alone), the *non-degenerate* ground-state entropy is identically zero (Section II; degenerate open-shell cases escape, Section II C). No transcendental enters.

- Under the metric embedding of $1/r_{12}$ onto the pair-state graph, entropy becomes finite. Diagonal pair entries $V_{ii}$ remain rational (Section V, [17]); spectral eigenvalues of $V_{ee}$ on the pair graph are irrational, consistent with the embedding-class signature of $1/r_{12}$ in Paper 18.

Entropy is therefore an embedding-class observable in the Paper 18 taxonomy. The combinatorial graph is entropy-blind (in the non-degenerate case); entropy is the price of projecting onto the metric continuum.

This placement is consistent with Paper 24's HO $\pi$-freeness rigidity theorem (the surviving spectral statement, not the retracted entropy corollary) [11]: the Bargmann–Segal harmonic oscillator lattice is bit-exactly $\pi$-free in rational arithmetic, and its one-body Hamiltonian has the same entanglement-inert character as $H_1$ here (it is one-body by construction). That $\pi$-freeness concerns the graph's eigenvalues and adjacency weights and is unaffected by the retraction in Section VII A; what does *not* follow from it — and what this paper previously asserted — is a corresponding rigidity of the interacting two-fermion *state*.

## VII. RIGIDITY PREDICTIONS: ONE RETRACTED, ONE SURVIVING

This section reports the outcome of the paper's two original predictions. Section VII A RETRACTS the first: the claimed rigidity theorem for the Bargmann–Segal harmonic-oscillator lattice — zero two-fermion closed-shell ground-state entropy for any central two-body potential — is false, and its supporting mechanism was a misreading of the Moshinsky–Talmi decomposition. Section VII B stands: it characterizes a universal two-variable scaling for Coulomb-lattice block entropy, asymptotic to second-order Rayleigh–Schrödinger perturbation theory. Only one of the two predictions therefore survives, and the surviving one is the Coulomb-side scaling, not the HO-side rigidity.

### A. HO zero-entropy theorem for closed-shell two-fermion ground states — retracted

**[RETRACTED 2026-08-22.]** This section previously reported prediction EP-2b as resolved in the strong form $S_{\rm HO} = 0$ *identically*, with the mechanism that on the Bargmann–Segal HO basis the one-body operator $H_{HO}$ commutes exactly with *any* central two-body interaction $V(r_{12})$, because the Moshinsky–Talmi transformation makes the total HO quanta $N_{\rm tot} = N_{\rm rel} + N_{CM}$ a conserved quantum number of $V$. **The claim, the mechanism, and the INTERNAL THEOREM tier are all withdrawn.**

*Why the mechanism is false.* The Moshinsky–Talmi *bracket* conserves total quanta — that is a property of the harmonic-oscillator coordinate transformation, and it is the standard nuclear-structure fact the section cited. The potential *matrix element* does not inherit it. A central $V(r_{\rm rel})$ is diagonal in the centre-of-mass quantum numbers and in $l_{\rm rel}$, but it *couples different relative-$n$*. Writing the element out, the bra bracket forces $2n + l + 2N_{CM} + \Lambda = N_{\rm bra}$ and the ket bracket forces $2n' + l + 2N_{CM} + \Lambda = N_{\rm ket}$, so

$$n - n' \;=\; \tfrac{1}{2}\,(N_{\rm bra} - N_{\rm ket}). \tag{8}$$

The $N_{\rm tot}$-changing elements are therefore exactly the $n \neq n'$ ones — the radial off-diagonal of $V$ in the relative coordinate — and they are nonzero. For the Minnesota singlet at $b = 1$, $\langle n_{\rm rel}{=}0, l{=}0|V|n_{\rm rel}{=}1, l{=}0\rangle = +17.3$ MeV against a diagonal $\langle 0, 0|V|0, 0\rangle = -0.55$ MeV (both at the $b$=1 relative-frame convention of the backing test; corrected 2026-08-29 — an earlier version printed $-0.81$,

which reproduces under no convention of the production Minnesota code). A magnitude caveat, same date: the $b$=1 contrast is convention-sensitive (the diagonal sits near a zero crossing); at the production convention ($b = 2.0364$, $\hbar\omega = 10$ MeV) the coupling is 0.48× the diagonal. The load-bearing evidence is the convention-independent commutator (0.74) and the nonzero corrected entropies, not the ratio.

The paper's own contrast should have caught this. The Interpretation paragraph argued that the Coulomb kernel is different because it "does not conserve any HO-like $N_{\rm tot}$ quantum number" — but $1/r_{12}$ *is* a central potential, so it falls under the same "any central $V(r_{12})$" the theorem quantified over. The two statements cannot both hold, and the contradiction was visible in the text.

*Where the zero came from.* The reported $S_{\rm full} = 0$ and $\|[H_{HO}, V]\|_F/\|H_{HO}\|_F \sim 10^{-16}$ were produced by an undocumented guard in the two-body matrix-element routine, which returned zero whenever the bra and ket total quantum numbers differed — that is, it discarded precisely the couplings identified by (8). With the guard removed, the same driver on the same basis gives Table II.

TABLE II. EP-2b CORRECTED: two-fermion HO on the Bargmann–Segal lattice, Minnesota $S = 0$ interaction at $\hbar\omega = 10$ MeV. $E_{\rm full}$ is the FCI ground-state energy, $E_{\rm kin}$ the noninteracting HO ground state, $S_{\rm full}$ the von Neumann entropy of the normalized (trace-one) spatial 1-RDM in nats, and $n_0$ the leading natural-orbital occupation (of a possible 2). Compare the retracted values: $E_{\rm full}$ was reported as 22.185 at every $N_{\rm max}$, $S_{\rm full}$ as 0.0, and the commutator ratio as $\sim 10^{-16}$.

| $N_{\rm max}$ | $E_{\rm full}$ (MeV) | $E_{\rm kin}$ (MeV) | $S_{\rm full}$ (nats) | $\|[H_{HO}, V]\|_F/\|H_{HO}\|_F$ | $n_0$ |
|---|---|---|---|---|---|
| 2 | 21.6538 | 30.000 | 0.0671 | 0.74 | 1.9800 |
| 3 | 21.6279 | 30.000 | 0.0716 | 0.63 | 1.9791 |
| 4 | 21.5442 | 30.000 | 0.0833 | 0.67 | 1.9764 |

*What is true instead.* The HO two-fermion closed-shell ground state is *not* a single Slater determinant for a central two-body interaction, and $H_{HO}$ does *not* commute with $V$ — the commutator is $O(1)$ relative to $\|H_{HO}\|_F$, not at the noise floor. The entropy is small (a few hundredths of a nat, comparable in magnitude to the $S \sim 4 \times 10^{-2}$ nats of the Coulomb EP-1 measurement) and it *grows* with basis size rather than staying pinned at zero. So the original heuristic that the section explicitly rejected as too weak — "a short-range kernel suppresses cusp concentration," giving $S_{\rm HO}$ small but nonzero rather than $S_{\rm HO} = 0$ — is closer to the truth than the structural statement that replaced it, though even that is not established here: at these basis sizes the HO and Coulomb entropies are the same order of magnitude.

*What survives.* The operator-theoretic result of Section II is untouched: it concerns the entanglement-inertness of the one-body operator and makes no claim about $N_{\rm tot}$. The Paper 24 $\pi$-freeness certificate is untouched: it concerns the graph's eigenvalues and adjacency weights, not the interacting two-fermion state. What is withdrawn is the bridge between them — the claim that a $\pi$-free graph certificate extends to a $\pi$-free (indeed vanishing) ground entropy — together with the asserted structural duality between this section and the nonzero Coulomb scaling of Section VII B. The two lattices are not opposites on this axis; on present evidence they behave similarly.

*Methodological note.* The corroborating evidence was itself a symptom. The result read as robust partly because $E_{\rm full}$ came out independent of $N_{\rm max}$ (both entries of the retracted table read 22.185), and the text presented that invariance as confirmation that "enlarging the single-particle basis does not mix higher shells into the ground state." That invariance was the guard, freezing the ground state inside a single $N_{\rm tot}$ block. Under the corrected computation $E_{\rm full}$ decreases monotonically with basis size, as a variational calculation must. A quantity that comes out *exactly* zero, or *exactly* invariant, where the physics predicts only "small" or "weakly dependent," is a reason to ask what has been restricted.

*Implementation.* The corrected measurements come from the production module `geovac.nuclear.ho_two_fermion` composed with `geovac.nuclear.moshinsky` and `geovac.nuclear.minnesota` from Paper 23 [10]. The retraction is pinned by `tests/test_paper27_entropy.py` and, on the Paper 24 side, by `tests/test_paper24_ho_entropy.py`, which additionally carries a tripwire reproducing the retired guard on demand so that a silent reinstatement fails the suite.

### B. Universal scaling of Coulomb block entropy (verified, reframed)

*Main result.* Block ground-state entropy on the Coulomb lattice obeys a universal two-variable scaling in the dimensionless off-diagonal $V_{ee}$ mass $\tilde{w}_B \equiv \|V_{ee}^{\rm off-diag(H_1)}\|_F/\|H_1\|_F$ and the dimensionless first $H_1$ gap $\delta_B \equiv \Delta E_1/\|H_1\|_F$:

$$S_B \;=\; A\,(\tilde{w}_B/\delta_B)^{\gamma}. \tag{9}$$

The local exponent approaches a basis-extension limit $\gamma_\infty \approx 1.96$–$1.97$ on the physical Hamiltonian (see the correction paragraph below). On the hop-augmented model it was extrapolated to 1.93–1.96 from the $n_{\rm max} = 2, 3, 4, 5$ local slopes at $Z = 15 \to 30$ $(1.992, 1.976, 1.966, 1.959)$; the model figures in the rest of this paragraph are retained as such. The precise limit is extrapolation-order dependent: a first-order Richardson / linear-in-$1/n_{\rm max}$ fit gives 1.93–1.94, Aitken $\Delta^2$ gives 1.93–1.95 (three-point-window dependent), and a linear-in-$1/n_{\rm max}^2$ fit gives 1.96. (The earlier text quoted 1.96 alone without stating the assumed order; corrected 2026-08-28.) The conclusion below 2 is method-independent — every extrapolation tried lands under 2, as does every tabulated slope. This is *below* the non-degenerate second-order

Rayleigh–Schrödinger prediction $\gamma = 2$. The persistent gap of $\sim 0.04$–$0.07$ at the basis limit is an aggregation effect: with more orbitals available, $S_B$ accumulates contributions from many small admixtures rather than being dominated by the lowest excitation, and the aggregate sum scales with a slightly smaller exponent. Finite-window fits give $\gamma \in [2.1, 2.7]$, depending monotonically on both $Z$ and $n_{\max}$. Across 22 calibration points (He-like $Z \in [2, 100]$ at $n_{\max} = 3$, plus a LiH bond-block $R$-sweep from 1.5 to 8 bohr) Eq. (9) fits at $R^2 = 0.990$ with $\gamma = 2.228$, $A = 0.157$; the only points with $> 25\%$ deviation are the dissociation-limit $R \in \{6, 8\}$ bohr LiH-bond points that saturate at $S_B = \log 2$ (the two-site ceiling).

*Correction (2026-09-25): which $H_1$.* Every He-like number in this subsection was measured with the hop-augmented $H_1$ of §II (the LiH bond-block points use the composed builder's $H_1$ and are unaffected). Re-measured with the physical $H_1$ of Eq. (3): (i) at $n_{\max} = 3$, $Z \in \{2, 3, 4, 6, 8, 10\}$, $S_B = 3.77 \times 10^{-2}$, $1.05 \times 10^{-2}$, $4.78 \times 10^{-3}$, $1.78 \times 10^{-3}$, $9.37 \times 10^{-4}$, $5.82 \times 10^{-4}$; $\delta_B = 0.2028$ exactly ($Z$-independent, since $H_1 \propto Z^2$) and $\tilde{w}_B \propto 1/Z$ exactly (from 0.0877 at $Z = 2$), so the single-variable fit and the $Z$-fit coincide: $\gamma = -\alpha_Z = 2.565$, $A = 16.5$, $R^2 = 0.995$ (model: $\gamma = 2.383$, $A = 8.16$, $\alpha_Z = -2.613$). The hop had been lowering the small-$Z$ slope; the physical $\gamma_{\rm loc}(Z{=}2)$ is 3.16 (model 2.64). (ii) The large-$Z$ local slopes ($Z : 15 \to 30$) are hardly moved, as expected since the hop is $Z$-independent while $H_1 \propto Z^2$: 1.992, 1.979, 1.974, 1.971, 1.970, 1.969 at $n_{\max} = 2$–7, and $\gamma_{\rm loc}(Z{:}50 \to 100) \approx 1.91$ at every $n_{\max}$. Extrapolating the physical local slopes gives $\gamma_\infty \approx 1.96$–1.97 (linear in $1/n_{\max}$: 1.964; linear in $1/n_{\max}^2$: 1.967; Aitken on $n_{\max} = 5, 6, 7$: 1.968). The structural conclusion is unchanged—every physical local slope at large $Z$ and every extrapolation lies below 2—and the extrapolated value moves up by $\approx 0.01$–$0.03$. (iii) The global-window exponents over $Z \in [2, 30]$ are higher on the physical Hamiltonian (Table IV). The paragraphs and tables below give the model values, retained for the record and labelled as such.

*Numerical confirmation (hop-augmented model).* Three calibrations pin Eq. (9). (i) He-like isoelectronic sequence at $n_{\max} = 3$, $Z \in \{2, \ldots, 10\}$ gives $\gamma = 2.383$, $A = 8.16$, $R^2 = 0.998$ in the single-variable form $S_B = A\tilde{w}_B^{\gamma}$; this is equivalent to Eq. (9) because $\delta_B$ is essentially constant at 0.20 across the He-like family. The companion $Z$-scaling fit $S_B \sim Z^{\alpha_Z}$ gives $\alpha_Z = -2.613$, $R^2 = 0.995$, consistent with Paper 26's single-orbital entropy $\alpha_Z = -2.56$ [12]. (ii) Extending to $Z \in \{12, 15, 20, 30, 50, 100\}$ gives a monotonically decreasing local log-log slope: $\gamma_{\rm loc}(Z = 2) = 2.64 \to \gamma_{\rm loc}(Z = 100) = 1.92$, crossing 2 between $Z = 10$ and $Z = 12$. The underlying $S_B(\tilde{w}_B/\delta_B)$ curve is smooth, not a pure power law; local linearization over any finite window picks up a slope that depends on the window. (iii) Fixing the Z-window $Z \in \{2, \ldots, 30\}$ and sweeping $n_{\max} \in \{2, 3, 4\}$ gives local slopes at large Z of 1.992, 1.976, and 1.966 respectively: these lie *below* 2 and decrease further with both $Z$ and $n_{\max}$, i.e. they move away from 2 from below. (It is the *global*-window exponents that approach 2 from above: $2.373, 2.223, 2.182$ over this $Z \in [2, 30]$ window (Table IV); $2.63, 2.38, 2.32$ over the $\{2, 3, 4, 6, 10\}$ window of robustness item (c).)

TABLE III. EP-2c pilot data, *hop-augmented model* (helium-like physical values are given in the 2026-09-25 correction paragraph of §VII B): He-like isoelectronic sequence at $n_{\max} = 3$. Full data in `debug/data/ep2c_entropy_vs_vee_offdiag.json`. $\delta_B \approx 0.20$ (constant to 1%) across the entire table.

| $Z$ | $S_B$ (nats) | $\tilde{w}_B$ |
|---|---|---|
| 2 | $4.08 \times 10^{-2}$ | 0.1052 |
| 3 | $1.12 \times 10^{-2}$ | 0.0648 |
| 4 | $4.98 \times 10^{-3}$ | 0.0470 |
| 6 | $1.82 \times 10^{-3}$ | 0.0306 |
| 8 | $9.48 \times 10^{-4}$ | 0.0226 |
| 10 | $5.86 \times 10^{-4}$ | 0.0179 |

*Methodological robustness.* Three potential confounders were checked and did not affect the main result. (a) The apparent 40% prefactor offset between "core" and "lone-pair" blocks in an earlier 16-block multi-block fit is a pure Z-range artifact; repeating the same fit on the *identical* He-like family restricted to the core and lone-pair Z-sets reproduces the offset without any block-type variation. The underlying curve is not a pure power law, and log–log regression over different Z-windows picks up different local slopes (finite-basis residue at small Z inflates $\gamma$). (b) A bounded ansatz $S_B = \log 2 \cdot \tanh^2(c\,(\tilde{w}_B/\delta_B)^{\gamma})$, motivated by the two-site $\log 2$ ceiling, fits markedly worse than the pure power law ($R^2 = 0.37$ vs. 0.90 on a 21-point mixed-geometry set) because the tanh Taylor expansion forces the small-argument slope to $2\gamma$ with a small prefactor $c^2$, incompatible with the $\sim 4$-decade dynamic range in $S_B$. Three additional bounded ansatze (rational saturation, stretched exponential, Hill function) were tested in log-space on a combined 32-point dataset (EP-2g + EP-2k HF and $H_2O$ bonds) and all give $R^2_{\log} \in [0.985, 0.989]$, statistically indistinguishable from the pure power law's 0.9895. The saturation at $\log 2$ is *discrete*, not smooth: only one or two LiH dissociation-limit points actually reach the ceiling, so saturation parameters cannot earn their fit cost. The pure power law Eq. (9) remains the best single functional form. (c) The $n_{\max} = 2, 3, 4$ sequence of global-window exponents $(\gamma, A) = (2.63, 0.73), (2.38, 0.18), (2.32, 0.09)$ shows monotone approach to $\gamma = 2$ from above, at fixed Z-window $\{2, 3, 4, 6, 10\}$; the local slope at the large-Z edge ($Z = 15 \to 30$) instead sits below 2 throughout and descends further (Table IV).

*Heavy-atom R-independence (EP-2k).* HF and $H_2O$ bond blocks were R-swept across $R \in [1.3, 3.0]$ bohr (HF)

TABLE IV. $\gamma(Z, n_{\max})$ surface (EP-2i + EP-2L). The local slope of $S_B$ vs. $\tilde{w}_B/\delta_B$ between $Z = 15$ and $Z = 30$ decreases monotonically with $n_{\max}$ and lies *below* 2 at every $n_{\max}$ tabulated. (The global-window column in contrast sits above 2 and descends toward it; the two columns move oppositely and should not be conflated.) Richardson extrapolation gives an asymptotic slope $\gamma_\infty \approx 1.96$–$1.97$ on the physical $H_1$ ($n_{\max} = 2$–$7$) and 1.93–1.96 on the hop-augmented model ($n_{\max} = 2$–$5$), extrapolation-order dependent, in every case *below* the non-degenerate Rayleigh–Schrödinger value of 2. The global-window $\gamma$ over $Z \in \{2, 3, 4, 6, 10, 15, 30\}$ is dominated by small-$Z$ finite-basis residue and is consistently higher.

| | physical $H_1$ (2026-09-25) | | hop-augmented model | |
|---|---|---|---|---|
| $n_{\max}$ | $\gamma$ (global) | $\gamma_{\text{loc}}$ $(15 \to 30)$ | $\gamma$ (global) | $\gamma_{\text{loc}}$ $(15 \to 30)$ |
| 2 | 2.400 | 1.992 | 2.373 | 1.992 |
| 3 | 2.321 | 1.979 | 2.223 | 1.976 |
| 4 | 2.293 | 1.974 | 2.182 | 1.966 |
| 5 | 2.280 | 1.971 | – | 1.959 |
| 6 | 2.273 | 1.970 | – | – |
| 7 | 2.268 | 1.969 | – | – |
| $\infty$ | — | $\approx 1.96$–$1.97$ | — | $\approx 1.93$–$1.96$ |

and $R_{OH} \in [1.5, 3.0]$ bohr ($H_2O$). Both are essentially *R-independent*: $S_B$ varies by 0.4% for HF and 0.2% for $H_2O$ across the entire range. At every R-point HF sits at $S_B/S_{\text{predicted}} = 1.20 \pm 0.02$ above Eq. (9), and $H_2O$ at $1.25 \pm 0.01$. In contrast, LiH varies by 70× across the same R-range and shows the saturation at $\log 2$ discussed above.

The mechanism is the Z asymmetry between center and partner. HF ($Z_F = 9$, $Z_H = 1$) and $H_2O$ ($Z_O = 8$, $Z_H = 1$) have bonds dominated by the heavy atom; the H-side orbital contributes a perturbative correction whose amplitude depends on R only through cross-center $V_{ne}$ matrix elements that vary slowly compared to the heavy-atom-dominated $H_1$ structure. LiH (Li-side $Z_{\text{eff}} = 1$ = H-side Z) is the symmetric special case where bond stretching genuinely populates two equal-charge centers, driving the quasi-degenerate regime and the eventual $\log 2$ saturation. For chemically realistic heavy-atom bonds at or near equilibrium, the universal curve holds within a 25% band that is R-independent within $\sim 1\%$.

*Bond-block qualifier.* For bond blocks with two-center $H_1$, $\delta_B$ is the *essential* second variable: the LiH bond-block point at equilibrium ($\tilde{w}_B = 0.042$) would give $S_B \approx 4 \times 10^{-3}$ under the single-variable fit but the actual value is 0.079, a 20× deviation. Including $\delta_B$ collapses it onto Eq. (9) within the universal band. Bond blocks in isolation, however, are not a single well-defined object: the composed LiH bond block (within-block Hamiltonian only, Li-side $Z_{\text{eff}} = 1$ = H-side $Z$) is *exactly degenerate* ($\delta_B = 0$) because the two sub-block 1s orbitals share the same effective charge. Only the cross-center $V_{ne}$ terms introduced by the balanced builder [9] break the degeneracy and put the bond block at a finite $(\tilde{w}_B, \delta_B)$. A similar mechanism explains why $BeH_2$ at equilibrium saturates at $\log 2$ in the balanced builder: cross-block ERIs from the Be-core and the second Be–H bond compress $\delta_B$ from 0.204 (within-block) to 0.027 (balanced), an 8× reduction that crosses into the saturation regime. By contrast, the $H_2O$ bond block shows no measurable shift ($S_{\text{balanced}}/S_{\text{composed}} = 1.002$), and its $1.25 \pm 0.01$ residual from the universal-curve prediction is intrinsic. The operational statement is: Eq. (9) applies when one chooses a specific $H_1$ (including its cross-block contributions) and tracks both $\tilde{w}_B$ and $\delta_B$ for that choice.

*Implementation.* Drivers: `debug/ep2c_entropy_vs_vee_offdiag.py` (single-block pilot), `ep2c_multi_block.py` (multi-block), `ep2d_zrange_control.py` (Z-range control), `ep2f_full_library.py` (library + high-Z), `ep2e_bond_block.py` (bond-block divergence), `ep2g_two_variable_scaling.py` (universal $w/\delta$ scaling), `ep2h_universality_extension.py` (other bonds + $n_{\max}$ + bounded-form tests), `ep2i_deep_diagnostics.py` ($BeH_2$ cross-block decomposition, $H_2O$ intrinsic offset, $\gamma(Z, n_{\max})$ surface), `ep2j_li_be_extension.py` (3e/4e atoms, non-degeneracy qualifier), `ep2k_heavy_atom_bond_sweep.py` (HF, $H_2O$ bond R-sweeps), `ep2L_nmax5_gamma.py` ($n_{\max} = 5$ asymptote and Richardson extrapolation), `ep2M_two_scale_bounded.py` (rational, stretched-exp, Hill bounded-form fits, all negative), `ep2N_be_analytical.py` (Be 3 × 3 degenerate-PT diagonalization). Data under `debug/data/ep2{c,d,e,f,g,h,i}_*.json`; fits regression-locked by `tests/test_paper27_entropy.py::test_paper27_ep2c_dimension` `::test_paper27_ep2c_multi_block_z_range_control`, `::test_paper27_ep2d_z_range_artifact`, `::test_paper27_ep2g_universal_w_over_delta`, `::test_paper27_ep2h_nmax_residue_persists`, and `::test_paper27_ep2i_gamma_asymptote`.

### C. Operational temperature decoding (Sprint TD Track 2)

The information-theoretic entropy of §II, $S_{\text{full}} = S_{\text{vN}}(\rho_1)$ of the bare-graph ground state, is the $T \to 0$ limit of the von Neumann entropy of the Gibbs spatial 1-RDM in standard quantum statistical mechanics. Sprint TD Track 2 (May 2026, `debug/sprint_td_track2_memo.md`) verifies this operationally on H, He, and $Li^+$ across 30 log-spaced temperatures spanning $0.005 \times \text{gap} \le k_B T \le 20 \times \text{gap}$, using the textbook Boltzmann-weighted ensemble over graph-native CI eigenstates. (These are eigenstates of the hop-augmented model $H_1$ of §II; the identity verified below is standard statistical mechanics and holds for any Hamiltonian, but the tabulated entropies are model values—helium's physical $S_{\text{full}}$ at $n_{\max} = 3$ is 0.03772, and $Li^+$'s is 0.01049.)

The thermodynamic entropy decomposes cleanly as

$$S_{\text{thermo}}(\beta) \;=\; k_B \cdot S_{\text{microstate info}}(\beta), \qquad (10)$$

where the right-hand side is the Shannon entropy of the microstate-resolved Boltzmann distribution. The identity holds at machine precision (max residual $4.8 \times 10^{-14}$ across all $3 \times 30 = 90$ test points). This is standard quantum statistical mechanics; the sprint's content is the explicit verification on graph-native CI states.

The von Neumann entropy of the Gibbs *spatial 1-RDM*, $S_{\text{vN}}(\rho_1(\beta))$ with $\rho_1(\beta) = \sum_n p_n(\beta)\,\rho_1^{(n)}$, is the "decoded-temperature spatial residual" — what remains of the information content after the apparatus parameter $\beta$ is factored out. At $T \to 0$ this residual equals $S_{\text{full}}(\text{GS})$ of §II to $\sim 3 \times 10^{-10}$ (not bit-exactly; the corpus reserves "bit-exact" for machine-precision skeleton identities):

TABLE V. Sprint TD Track 2: $T \to 0$ residual matches $S_{\text{full}}(\text{GS})$ of §II. Data `debug/data/sprint_td_track2.json`.

| System | $S_{\text{vN}}(\rho_1(\beta \to \infty))$ | $\lvert\Delta\rvert$ vs. $S_{\text{full}}(\text{GS})$ |
|---|---|---|
| H ($Z = 1$, $n_{\max} = 6$) | 0.000000 | 0 (single det. rigidity) |
| He ($Z = 2$, $n_{\max} = 3$) | 0.040811 | $3.1 \times 10^{-10}$ |
| $Li^+$ ($Z = 3$, $n_{\max} = 3$) | 0.011212 | $3.2 \times 10^{-10}$ |

In the $T \to \infty$ limit the residual approaches $\log(n_{\text{orbitals}}) = \log 14 \approx 2.639$ for both $N_e = 2$ systems, the standard Boltzmann thermal-mixing limit over the graph basis. Lindblad concavity bounds hold at every $\beta$ ($\max_n S_n \le S_{\text{vN}}(\rho_1(\beta)) \le S_{\text{apparatus}} + \sum_n p_n S_n$).

Operationally, this positions the Paper 27 information entropy as the dimensionless content of the GeoVac graph that remains after a finite-temperature observation projection (Paper 34 §III.15) is undone. Sprint TD Track 5 separately PSLQ-tested whether $S_{\text{full}}(\text{GS})$ values for He, $Li^+$, and He at $n_{\max} = 4$ sit in the master Mellin engine ring (Paper 32 §VIII case-exhaustion theorem and Paper 18 §III.7); the test returned a clean negative against a 12,312-form mechanical basis at 150 dps across all three systems, with control-class baseline matching to $z = 0$. Three orders of magnitude in coefficient relaxation ($10^3 \to 10^6$) does not change the verdict. GeoVac correlation entropy is genuinely von Neumann-only; it is not a master Mellin output. This is structurally distinct from spectral-action entropies (e.g., the Connes–Chamseddine area term on Euclidean Schwarzschild, which lives in $M_2$). $S_{\text{full}}$ joins $K = \pi(B + F - \Delta)$, the Sprint MR-C $L_2$ next-order constant $c \approx 4.10932\ldots$, and the Wolfenstein parameters of Sprint W3 on the list of natural framework constants outside the master Mellin ring.

We do not speculate about gravitational, holographic, or black-hole entropy in this paper. Whether the pair-counting re-interpretation of the area law in Section IV admits a continuum geometric reading is a separate question; the operator-theoretic content stands without it.

## VIII. CONCLUSION

Five results have been established on GeoVac lattices for closed-shell two-fermion ground states. First, the von Neumann entropy of any single-particle reduced density matrix derived from the *non-degenerate* ground state of the one-body Hamiltonian alone is identically zero; numerically, $S_{\text{kin}}/S_{\text{full}} \sim 10^{-14}$ at $n_{\max} = 2, 3$ on a He model whose one-body operator is deliberately non-diagonal (a graph hop, which is not part of helium's Hamiltonian; with helium's physical, diagonal $H_1$ the zero is exact). All ground-state entanglement is generated by the two-body operator $V_{ee}$. Second, the area-law scaling $S_n \propto 4 \log n$ asserted on one-particle grounds in Paper 5 [5] is recovered as a two-body pair-counting statement $A_n = g_n^2 \propto n^4$, with the factor of four as the two-body signature. Third, the cusp localizes as a single hot node on the $V_{ee}$ pair graph, making the loci of energy correlation, entropy generation and cusp mass coincident — established by an edge-resolved entropy decomposition on the pair-state graph, not inferred. Fourth — and negatively — the claim that two-fermion closed-shell ground-state entropy on the Bargmann–Segal harmonic oscillator lattice is identically zero for *any* central two-body potential is RETRACTED (§VII A). The Moshinsky–Talmi *bracket* conserves total oscillator quanta; the potential *matrix element* does not, and the published zero was an artifact of a code guard that discarded exactly the couplings which change them. Corrected, the entropy is nonzero, comparable in magnitude to the Coulomb value, and increasing with basis size. Fifth, on the Coulomb lattice, block entropy obeys a universal two-variable scaling $S_B = A\,(\tilde{w}_B/\delta_B)^{\gamma}$ in the dimensionless $V_{ee}$ off-diagonal mass and the first $H_1$ gap, with the local exponent extrapolating to $\gamma_\infty \approx 1.96$–$1.97$ as $Z, n_{\max} \to \infty$ on the physical Hamiltonian – below the naive second-order Rayleigh–Schrödinger value of 2, by a residue attributed to multi-shell aggregation in the basis-extension limit.

These results place entropy in the exchange-constant taxonomy of Paper 18 [8] as a projection-class observable: zero on the combinatorial graph (non-degenerate case), finite only after the metric embedding of $1/r_{12}$. The two-body character of the area-law scaling is structural, not incidental; on present evidence the Coulomb and harmonic oscillator two-fermion lattices behave similarly on this axis (the sharp-contrast reading is retracted, v5.0.0).

## EQUATION VERIFICATION

Per Section 13.4a of the project guidelines, every numerical claim in this paper is traceable to a debug data file or to a test in the codebase.

- Physical-Hamiltonian re-measurements of 2026-09-25 (Table I physical rows, Section V diagonal fraction and commutator, the §VII B correction paragraph's EP-2c fit): **Verified** by

`tests/test_he_physical_hamiltonian.py`; the hop-augmented model is kept distinct from helium by `tests/test_he_hop_guard.py`. The items below verify the model values.

- Table I: `debug/data/ep1_entropy_partition.json`. **Verified** by `tests/test_paper27_entropy.py::test_paper27_table1_ep1_reproduction`, which re-runs the EP-1 pipeline (FCI of $H_1$ vs. $H_1 + V_{ee}$ on the He angular momentum basis, via `geovac/casimir_ci.py` and `geovac/hypergeometric_slater.py`) and checks $|S_{\rm kin}| < 10^{-12}$ and $|S_{\rm kin}|/S_{\rm full} < 10^{-10}$ at $n_{\rm max} \in \{2, 3\}$.

- Eq. (4)–(6): combinatorial identities on Paper 0 shell degeneracies $g_n = 2n^2$. **Verified** symbolically by `tests/test_paper27_entropy.py::test_paper27_area_law_pair_counts` and `::test_paper27_area_law_leading_order_agrees`, with companion test `::test_paper27_one_body_counts_differ_from_area_law` confirming the one-particle slope is 2 (single-shell) or 3 (cumulative), distinct from the two-body slope 4. Hand-tabulation in EP-3 analysis [16].

- Section V numerical values ($\|V_{ee}^{\rm diag\,(H_1)}\|_F/\|V_{ee}\|_F$, relative commutator, hot-node weights): produced by `debug/energy_graph_exploration.py`, data in `debug/data/energy_graph_nmax{3,4}.json`. **Verified** by `tests/test_paper27_entropy.py::test_paper27_energy_graph_structural_invariants` (commutator ratio, diagonal fraction, and exact $V_{(1s,1s),(1s,1s)} = 5/8$ at both $n_{\rm max} = 3$ and 4); non-vanishing of the commutator in the $n_{\rm max} = 3 \to 4$ extension is pinned by `::test_paper27_commutator_saturates_not_vanishes` (`--slow`).

- Table III and Eq. (9) (the dimensionless power law $S_B = 8.16\,\tilde{w}_B^{2.383}$, $R^2 = 0.998$): **Verified** by `tests/test_paper27_entropy.py::test_paper27_ep2c_dimensionless_power_law`, data in `debug/data/ep2c_entropy_vs_vee_offdiag.json`.

- Table II and Section VII A corrected numerical claims (EP-2b RETRACTED v5.0.0: $S_{\rm HO}$ small but nonzero, 0.0671/0.0716/0.0833 nats, natural-orbital occupation $\approx (1.98, \ldots)$, commutator $O(1)$): **Verified** by `tests/test_paper27_entropy.py::test_paper27_ep2b_ho_gs_is_not_a_single_determinant` and `::test_paper27_ep2b_ho_kinetic_interaction_does_no` data in `debug/data/ep2b_ho_two_fermion.json`.

- Section V entropy–cusp co-location (58.5% of $S$ on the hottest $V_{ee}$ edge at $n_{\rm max} = 3$, 54.1% at $n_{\rm max} = 4$; zero from a comparably large non-reference edge; Spearman $\rho = +0.87$; re-measured 2026-09-25 on helium's physical Hamiltonian): **Verified** by `tests/test_paper27_entropy_locus.py`.

- Section VI embedding-class placement: diagonal pair-state $V_{ii}$ rationality and spectral irrationality verified in EP-3 / energy-graph analysis [17]; consistent with Paper 18 classification.

The operator-theoretic statement of Section II is a known free-fermion fact [1]; no new verification is required beyond the numerical confirmation in Section III.

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